

Learning Green's Functions as Operator Kernels for Linear Boundary Value Problems

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Abstract

Green's functions provide an explicit representation of inverse linear operators associated with differential equations and boundary conditions. While classical numerical methods compute solutions for specific forcing terms, learning the Green's function itself enables a compact and interpretable representation of the underlying operator. In this work, we investigate the feasibility of learning Green's functions from input-output data using constrained machine learning models. Focusing on the one-dimensional Poisson equation with Dirichlet boundary conditions, we frame Green's function learning as an operator learning problem and analyze the role of physics-based structural constraints such as symmetry, boundary conditions, and regularity. This project serves as a controlled study of kernel learning for linear physical systems and as a baseline for future extensions to higher-dimensional and parameterized operators.

1 Introduction and Motivation

Linear differential operators appear ubiquitously in physics, governing phenomena such as electrostatics, diffusion, elasticity, and quantum mechanics. For a given linear operator and boundary conditions, the solution to a boundary value problem is fully characterized by the corresponding Green's function, which represents the kernel of the inverse operator.

Classical numerical methods solve for the system response to a specific forcing function, whereas the Green's function encodes the response to *all* admissible forcings. Learning the Green's function, therefore, constitutes a stronger and more interpretable objective than learning individual solutions.

Recent developments in machine learning have focused on learning solution maps for parametric partial differential equations. In contrast, this work explores the learning of the operator kernel itself, explicitly incorporating physical structure into the learning process.

2 Mathematical Background

Consider a linear boundary value problem

$$Lu(x) = f(x), \quad x \in \Omega, \tag{1}$$

where L is a linear differential operator acting on functions defined on a domain Ω , subject to appropriate boundary conditions.

If the inverse operator L^{-1} exists, the solution may be written in integral form:

$$u(x) = \int_{\Omega} G(x, x') f(x') dx', \tag{2}$$

where $G(x, x')$ is the Green's function associated with L and the imposed boundary conditions.

The Green's function is thus the integral kernel of the inverse operator and fully encodes the system's response characteristics.

3 Problem Formulation

3.1 Physical setting

We focus on the one-dimensional Poisson equation with Dirichlet boundary conditions:

$$-\frac{d^2 u}{dx^2} = f(x), \quad x \in [0, 1], \quad (3)$$

subject to

$$u(0) = u(1) = 0. \quad (4)$$

The forcing function $f(x)$ is assumed to lie in $L^2([0, 1])$.

3.2 Green's function representation

The corresponding Green's function satisfies

$$-\frac{d^2}{dx^2} G(x, x') = \delta(x - x'), \quad (5)$$

with boundary conditions

$$G(0, x') = G(1, x') = 0. \quad (6)$$

The analytic Green's function is given by

$$G(x, x') = \begin{cases} x(1 - x') & x \leq x', \\ x'(1 - x) & x > x'. \end{cases} \quad (7)$$

3.3 Learning objective

Rather than learning the solution map $f \mapsto u$, we aim to learn an approximation to the Green's function itself:

$$\hat{G}_\theta : [0, 1] \times [0, 1] \rightarrow \mathbb{R}, \quad (8)$$

such that

$$u(x) \approx \int_0^1 \hat{G}_\theta(x, x') f(x') dx'. \quad (9)$$

This corresponds to learning an explicit approximation of the inverse operator L^{-1} .

4 Structural Constraints

The Green's function satisfies several structural properties:

Symmetry Since the operator is self-adjoint,

$$G(x, x') = G(x', x). \quad (10)$$

Boundary conditions

$$G(0, x') = G(1, x') = 0. \quad (11)$$

Regularity The Green’s function is smooth for $x \neq x'$, with a discontinuity in its first derivative at $x = x'$.

These properties are enforced during training via constraint terms in the loss function.

5 Data Generation

Training data consists of pairs (f_i, u_i) generated as follows:

- Sample forcing functions $f_i(x)$ from a predefined function family (e.g. Fourier series with random coefficients or Gaussian random fields).
- Compute solutions $u_i(x)$ using analytic convolution with the true Green’s function or high-resolution numerical solvers.

The Green’s function itself is not provided to the model during training.

6 Baseline Methods

The learned Green’s function is compared against the following baselines:

- Finite difference solvers based on matrix inversion.
- Spectral methods using Fourier representations.
- Kernel ridge regression treating the solution operator as a black-box kernel.

These baselines establish numerical accuracy and highlight the trade-offs between classical solvers and learned operator representations.

7 Learning Methodology

A parameterized model $\hat{G}_\theta(x, x')$ is trained by minimizing the loss

$$\mathcal{L}(\theta) = \mathbb{E}_f \left[\left\| u_{\text{true}} - \int_0^1 \hat{G}_\theta(x, x') f(x') dx' \right\|_{L^2}^2 \right] + \lambda \mathcal{R}_{\text{physics}}, \quad (12)$$

where $\mathcal{R}_{\text{physics}}$ enforces symmetry, boundary conditions, and smoothness.

The focus is on interpretability and physical consistency rather than architectural complexity.

8 Evaluation Criteria

Model performance is evaluated using:

- L^2 error between predicted and true solutions.
- Error between learned and analytic Green’s functions.
- Generalization to unseen forcing functions.
- Stability under perturbed or noisy inputs.

9 Scope and Limitations

This study is restricted to linear operators on fixed domains with prescribed boundary conditions. No claims are made regarding computational efficiency relative to classical numerical solvers.

The primary objective is to establish a clean and interpretable framework for learning operator kernels in physics-informed settings.

10 Future Extensions

Natural extensions include:

- Higher-dimensional domains.
- Parameterized operators.
- Time-dependent Green’s functions.
- Nontrivial geometries and boundary conditions.

These extensions are left for future work.